

Methods

Table 1: Simulation mis-specification design matrix. Outer dimensions describe the unique correct models (6) while inner dimensions (18) describe the unique mis-specifications run for each simulation.

Covariance Matrix	Autocorrelation	Matern Correlation	Phylogenetic Correlation
Type	Mis-specification		
LMM	Data Model	Data Model	Data Model
	iid	iid	iid
	Mis-specified Covariance	Mis-specified Covariance	Mis-specified Covariance
GLMM	Data Model	Data Model	Data Model
	iid	iid	iid
	Mis-specified Covariance	Mis-specified Covariance	Mis-specified Covariance

Table 2: Linear Model Simulation: data generating models, parameter values, and mis-specifications.

Data Generating Model		Parameters	Data Fitting Model
Correct	$X_i \sim N(0, 1)$	$\beta = (4, -5)$ $\sigma_y = 1$	$X_i \sim N(0, 1)$
	$\mu_{i,j} = X_i \beta$		$\mu_{i,j} = X_i \beta$
	$y_{i,j} \sim N(\mu_{i,j}, \sigma_y)$		$y_{i,j} \sim N(\mu_{i,j}, \sigma_y)$
Mis-specified	$X_i \sim N(0, 1)$		$X_i \sim N(0, 1)$
	$\mu_{i,j} = X_i \beta$		$\mu_{i,j} = X_i \beta$
	$y'_{i,j} \sim N(\mu_{i,j}, exp(\sigma_y))$		$y'_{i,j} \sim N(\mu_{i,j}, \sigma_y)$

Linear Mixed Model			Generalized Linear Mixed Model			
	Data Generating Model	Parameters	Data Fitting Model	Data Generating Model	Parameters	Data Fitting Model
Correct	$\mu_i = u_{i-1} + a$	$a = 2$	$\mu_i = u_{i-1} + a$	$\mu_i = u_{i-1} + a$	$a = .02$	$\mu_i = u_{i-1} + a$
	$u_i \sim N(\mu_i, \sigma_u)$	$u[1] = 0$	$u_i \sim N(\mu_i, \sigma_u)$	$u_i \sim N(\mu_i, \sigma_u)$	$u[1] = 0$	$u_i \sim N(\mu_i, \sigma_u)$
	$y_i \sim N(u_i, \sigma_y)$	$\sigma_u = 1$	$y_i \sim N(u_i, \sigma_y)$	$y_i \sim Gamma(\frac{1}{CV}^2, e^{u_i} CV^2)$	$\sigma_u = 0.1$	$y_i \sim Gamma(\frac{1}{CV}^2, e^{u_i} CV^2)$
		$\sigma_y = 1$			$CV = 0.3$	
Mis-specified	Missing Random Effects			Missing Random Effects		
	$\mu_i = u_{i-1} + a$			$\mu_i = u_{i-1} + a$		
	$u_i \sim N(\mu_i, \sigma_u)$		$y_i \sim N(a(1 : n), \sigma_y)$	$u_i \sim N(\mu_i, \sigma_u)$		$y_i \sim Gamma(\frac{1}{CV}^2, e^a CV^2)$
	$y_i \sim N(u_i, \sigma_y)$			$y_i \sim Gamma(\frac{1}{CV}^2, e^{u_i} CV^2)$		
	Mis-specified Data Model			Misp-specified Data Model		
	$\mu_i = u_{i-1} + a$			$\mu_i = u_{i-1} + a$		$\mu_i = u_{i-1} + a$
	$u_i \sim N(\mu_i, \sigma_u)$		$\mu_i = u_{i-1} + a$	$u_i \sim N(\mu_i, \sigma_u)$		$u_i \sim N(\mu_i, \sigma_u)$
	$\sigma_y^2 = c(rep(35, n/4), rep(0.5, n/4),$		$u_i \sim N(\mu_i, \sigma_u)$	$y_i \sim Gamma(\frac{1}{CV}^2, e^{u_i} CV^2)$		$y_i \sim N(u_i, \sigma_y)$
	$rep(35, n/4), rep(0.5, n/4)$		$y_i' \sim N(u_i, \sigma_y)$			
	$y_i' \sim N(u_i, \sigma_y)$					
Mis-specified Random Effects			Mis-specified Random Effects			
$\mu_i = u_{i-1} + a$		$\mu_i = u_{i-1}$	$u_i' = u_{i-1} + Gamma(0.5, 10)$		$\mu_i = u_{i-1} + a$	
$u_i \sim N(\mu_i, \sigma_u)$		$u_i \sim N(\mu_i, \sigma_u)$	$y_i' \sim Gamma(\frac{1}{CV}^2, e^{u_i'} CV^2)$		$u_i \sim N(\mu_i, \sigma_u)$	
$y_i \sim N(u_i, \sigma_y)$		$y_i \sim N(u_i, \sigma_y)$			$y_i' \sim Gamma(\frac{1}{CV}^2, e^{u_i} CV^2)$	

Table 4: Spatial Model Simulation: data generating models, parameter values, and mis-specifications.

Linear Mixed Model			Generalized Linear Mixed Model			
	Data Generating Model	Parameters	Data Fitting Model	Data Generating Model	Parameters	Data Fitting Model
Correct		$\beta_0 = 2$			$\beta = 2$	
	$\omega \sim GMRP(Q[\kappa, \sigma_\omega^2])$	$\sigma_y^2 = 1$	$\omega \sim GMRP(Q[\kappa, \sigma_\omega^2])$	$\omega \sim GMRP(Q[\kappa, \sigma_\omega^2])$	$\phi = 30$	$\omega \sim GMRP(Q[\kappa, \sigma_\omega^2])$
	$\eta_i = \beta_0 + \omega_i$	$\phi = 50$	$\eta_i = \beta_0 + \omega_i$	$\eta_i = exp(\beta_0 + \omega_i)$	$\kappa = \sqrt{8}/\phi$	$\eta_i = exp(\beta_0 + \omega_i)$
	$y \sim N(\eta, \sigma_y)$	$\kappa = \sqrt{8}/\phi$	$y \sim N(\eta, \sigma_y)$	$y \sim Pois(\eta, \sigma_y)$	$\sigma_\omega^2 = 0.25$	$y \sim Pois(\eta, \sigma_y)$
		$\sigma_\omega^2 = 2$				
Mis-specified	Missing Random Effects			Missing Random Effects		
	$\omega \sim GMRP(Q[\kappa, \sigma_\omega^2])$			$\omega \sim GMRP(Q[\kappa, \sigma_\omega^2])$		
	$\eta_i = \beta_0 + \omega_i$		$\eta_i = \beta_0$	$\eta_i = exp(\beta_0 + \omega_i)$		$\eta_i = exp(\beta_0)$
	$y \sim N(\eta, \sigma_y)$		$y \sim N(\eta, \sigma_y)$	$y \sim Pois(\eta)$		$y \sim Pois(\eta)$
	Mis-specified Data Model			Mis-specified Data Model		
	$\omega \sim GMRP(Q[\kappa, \sigma_\omega^2])$		$\omega \sim GMRP(Q[\kappa, \sigma_\omega^2])$	$\omega \sim GMRP(Q[\kappa, \sigma_\omega^2])$		$\omega \sim GMRP(Q[\kappa, \sigma_\omega^2])$
	$\eta_i = \beta_0 + \omega_i$		$\eta_i = \beta_0 + \omega_i$	$\eta_i = exp(\beta_0 + \omega_i)$		$\eta_i = exp(\beta_0 + \omega_i)$
	$y \sim N(\eta, \sigma_y)$		$\sigma_y = exp(N(0, 1))$	$y' \sim B(1, 0.7) * Pois(\eta)$		$y \sim Pois(\eta)$
			$y \sim N(\eta, \sigma_y)$			
	Mis-specified Random Effects Model			Mis-specified Random Effects Model		
$\omega \sim GMRP(Q[\kappa, \sigma_\omega^2])$		$\omega \sim GMRP(Q[\kappa, \sigma_\omega^2])$	$R = 45^\circ; S = \sqrt{(15)}$		$\omega \sim GMRP(Q[\kappa, \sigma_\omega^2])$	
$\eta_i = \beta_0 + exp(\omega_i)$		$\eta_i = \beta_0 + \omega_i$	$Loc_{aniso} = (Loc)S^{-1}R^{-1}$		$\eta_i = \beta_0 + \omega_i$	
$y' \sim N(\eta, \sigma_y)$		$y' \sim N(\eta, \sigma_y)$	$\omega \sim GMRP(Q[\kappa, \sigma_\omega^2])$		$y' \sim N(\eta, \sigma_y)$	
			$\eta_i = \beta_0 + \omega_i$			
			$y' \sim N(\eta, \sigma_y)$			

Table 5: Phylogenetic Model Simulation: data generating models, parameter values, and mis-specifications.

Linear Mixed Model				Generalized Linear Mixed Model			
	Data Generating Model	Parameters	Data Fitting Model	Data Generating Model	Parameters	Data Fitting Model	
Correct	$tree \sim randomTree(n)$		$tree \sim randomTree(n)$	$tree \sim randomTree(n)$		$tree \sim randomTree(n)$	
	$\Sigma = BM(tree, a, r, \sigma_u^2)$	$a = 0$	$\Sigma = BM(tree, a, r, \sigma_u^2)$	$\Sigma = BM(tree, a, r, \sigma_u^2)$	$a = 0$	$\Sigma = BM(tree, a, r, \sigma_u^2)$	
	$u \sim MVNORM(\Sigma)$	$r = 0$	$u \sim MVNORM(\Sigma)$	$u \sim MVNORM(\Sigma)$	$r = 0$	$u \sim MVNORM(\Sigma)$	
	$X_i \sim Unif(-0.5, 0.5)$	$\sigma_u^2 = 2$	$X_i \sim Unif(-0.5, 0.5)$	$X_i \sim Unif(-0.5, 0.5)$	$\sigma_u^2 = 1$	$X_i \sim Unif(-0.5, 0.5)$	
	$\eta = X\beta + u$	$\beta = (0, 1)$	$\eta = X\beta + u$	$\eta = exp(X\beta + u)$	$\beta = (0.5, -2)$	$\eta = exp(X\beta + u)$	
	$y \sim Normal(\eta, \sigma_y)$	$\sigma_y = 1$	$y \sim Normal(\eta, \sigma_y)$	$y \sim Poisson(\eta)$		$y \sim Poisson(\eta)$	
Mis-specified	Missing Random Effects			Missing Random Effects			
	$tree \sim randomTree(n)$			$tree \sim randomTree(n)$			
	$\Sigma = BM(tree, a, r, \sigma_u^2)$			$\Sigma = BM(tree, a, r, \sigma_u^2)$			
	$u \sim MVNORM(\Sigma)$		$X_i \sim Unif(-0.5, 0.5)$	$u \sim MVNORM(\Sigma)$		$X_i \sim Unif(-0.5, 0.5)$	
	$X_i \sim Unif(-0.5, 0.5)$		$\eta = X\beta$	$X_i \sim Unif(-0.5, 0.5)$		$\eta = exp(X\beta)$	
	$\eta = X\beta + u$		$y \sim Normal(\eta, \sigma_y)$	$\eta = exp(X\beta + u)$		$y \sim Poisson(\eta)$	
	$y \sim Normal(\eta, \sigma_y)$			$y \sim Poisson(\eta)$			
	Mis-specified Data Model			Misp-specified Data Model			
	$tree \sim randomTree(n)$		$tree \sim randomTree(n)$	$tree \sim randomTree(n)$		$tree \sim randomTree(n)$	
	$\Sigma = BM(tree, a, r, \sigma_u^2)$		$\Sigma = BM(tree, a, r, \sigma_u^2)$	$\Sigma = BM(tree, a, r, \sigma_u^2)$		$\Sigma = BM(tree, a, r, \sigma_u^2)$	
	$u \sim MVNORM(\Sigma)$		$u \sim MVNORM(\Sigma)$	$u \sim MVNORM(\Sigma)$		$u \sim MVNORM(\Sigma)$	
	$X_i \sim Unif(-0.5, 0.5)$		$X_i \sim Unif(-0.5, 0.5)$	$X_i \sim Unif(-0.5, 0.5)$		$\eta = exp(\beta_0 + u)$	
$\eta = X\beta + u$		$\eta = exp(X\beta + u)$	$\eta = exp(X\beta + u)$		$y \sim Poisson(\eta)$		
$y \sim Normal(\eta, \sigma_y)$		$y \sim Normal(\eta, \sigma_y)$	$y \sim Poisson(\eta)$				
	Mis-specified Random Effects			Mis-specified Random Effects			
	$tree \sim randomTree(n)$		$tree \sim randomTree(n)$	$tree \sim randomTree(n)$		$tree \sim randomTree(n)$	
	$\Sigma = BM(tree, a, r, \sigma_u^2)$		$\Sigma = BM(tree, a, r, \sigma_u^2)$	$\Sigma = EB(tree, a = 1, r = -2, \sigma_u^2)$		$\Sigma = BM(tree, a, r, \sigma_u^2)$	
	$u \sim MVNORM(\Sigma)$		$u \sim MVNORM(\Sigma)$	$u' \sim MVNORM(\Sigma)$		$u \sim MVNORM(\Sigma)$	
	$X_i \sim Unif(-0.5, 0.5)$		$X_i \sim Unif(-0.5, 0.5)$	$X_i \sim Unif(-0.5, 0.5)$		$X_i \sim Unif(-0.5, 0.5)$	
	$\eta = X\beta + exp(u)$		$\eta = X\beta + u$	$\eta' = exp(X\beta + u')$		$\eta = exp(X\beta + u)$	
	$y' \sim Normal(\eta, \sigma_y)$		$y' \sim Normal(\eta, \sigma_y)$	$y' \sim Poisson(\eta')$		$y' \sim Poisson(\eta)$	

Results

LM

Table 6: Linear Model. Type I error rates at the 0.05 significance level evaluated for each method for theoretical and estimated residuals.

test	residual type	Pearson	one-step cdf	one-step Generic	one-step Gaussian	full Gaussian	ecdf
Kolmogorov-Smirnov	theoretical	0.048	0.048	0.048	0.048	0.048	0.044
	estimated	0.000	0.000	0.000	0.000	0.000	0.000

Table 7: Linear Model. Power at the 0.95 significance level evaluated for each method for theoretical and estimated residuals.

test	residual type	Pearson	one-step cdf	one-step Generic	one-step Gaussian	full Gaussian	ecdf
Kolmogorov-Smirnov	theoretical	1.000	1.000	1.000	1.000	1.000	1.000
	estimated	0.963	0.963	0.963	0.963	0.963	0.961

Temporal Correlation - LMM

Table 8: LMM Temporal Correlation Model. Type I error rates and Power evaluated for each analytical and simulation method for theoretical residuals. Results are partitioned out by model mis-specification (from left to right) and residual type (top to bottom).

test	method	Type I Error	Power		
		Correct	A: iid	B: Heterosck.	C: Missing Drift
Anderson-Darling	Pearson	0.061	1.000	1.000	1.000
Anderson-Darling	one-step cdf	0.046	1.000	1.000	1.000
Anderson-Darling	one-step Generic	0.046	1.000	1.000	1.000
Anderson-Darling	one-step Gaussian	0.046	1.000	1.000	1.000
Anderson-Darling	full Gaussian	0.046	1.000	1.000	1.000
Anderson-Darling	MCMC	0.048	1.000	1.000	0.050
Anderson-Darling	ecdf	0.228	1.000	1.000	1.000
Kolmogorov- Smirnov	Pearson	0.056	1.000	1.000	1.000
Kolmogorov- Smirnov	one-step cdf	0.041	1.000	0.889	1.000
Kolmogorov- Smirnov	one-step Generic	0.042	1.000	0.867	1.000
Kolmogorov- Smirnov	one-step Gaussian	0.042	1.000	0.867	1.000
Kolmogorov- Smirnov	full Gaussian	0.042	1.000	0.867	1.000
Kolmogorov- Smirnov	MCMC	0.041	1.000	0.814	0.047
Kolmogorov- Smirnov	ecdf	0.057	1.000	1.000	1.000
Autocorrelation	Pearson	0.047	1.000	0.000	1.000
Autocorrelation	one-step cdf	0.051	1.000	0.000	0.100
Autocorrelation	one-step Generic	0.052	1.000	0.000	0.104
Autocorrelation	one-step Gaussian	0.052	1.000	0.000	0.100
Autocorrelation	full Gaussian	0.052	1.000	0.000	0.100
Autocorrelation	MCMC	0.048	1.000	0.000	0.069
Autocorrelation	ecdf	0.049	0.983	0.000	0.986

Table 9: LMM Temporal Correlation Model. Type I error rates and Power evaluated for each analytical and simulation method for estimated residuals. Results are partitioned out by model mis-specification (from left to right) and residual type (top to bottom).

test	method	Type I Error	Power		
		Correct	A: iid	B: Heterosck.	C: Missing Drift
Anderson-Darling	Pearson	0.002	1.000	0.010	0.000
Anderson-Darling	one-step cdf	0.030	1.000	0.016	1.000
Anderson-Darling	one-step Generic	0.024	1.000	0.016	0.995
Anderson-Darling	one-step Gaussian	0.022	1.000	0.014	0.996
Anderson-Darling	full Gaussian	0.027	1.000	0.022	0.997
Anderson-Darling	MCMC	0.053	1.000	0.117	0.051
Anderson-Darling	ecdf	0.010	1.000	0.672	0.000
Kolmogorov- Smirnov	Pearson	0.129	1.000	0.668	1.000
Kolmogorov- Smirnov	one-step cdf	0.000	1.000	0.362	1.000
Kolmogorov- Smirnov	one-step Generic	0.000	1.000	0.365	1.000
Kolmogorov- Smirnov	one-step Gaussian	0.000	1.000	0.365	1.000
Kolmogorov- Smirnov	full Gaussian	0.000	1.000	0.370	1.000
Kolmogorov- Smirnov	MCMC	0.054	1.000	0.467	0.051
Kolmogorov- Smirnov	ecdf	0.129	1.000	0.657	1.000
Lilliefors	Pearson	0.051	0.130	0.992	0.022
Lilliefors	one-step cdf	0.048	0.130	0.980	1.000
Lilliefors	one-step Generic	0.052	0.130	0.980	0.036
Lilliefors	one-step Gaussian	0.053	0.130	0.980	0.036
Lilliefors	full Gaussian	0.052	0.130	0.982	0.034
Lilliefors	MCMC	0.050	0.130	0.870	0.054
Lilliefors	ecdf	0.051	0.289	0.995	0.078
Autocorrelation	Pearson	0.001	1.000	0.000	0.000
Autocorrelation	one-step cdf	0.003	1.000	0.000	0.001
Autocorrelation	one-step Generic	0.003	1.000	0.000	0.000
Autocorrelation	one-step Gaussian	0.003	1.000	0.000	0.000
Autocorrelation	full Gaussian	0.003	1.000	0.000	0.000
Autocorrelation	MCMC	0.027	1.000	0.104	0.059
Autocorrelation	ecdf	0.001	1.000	0.048	0.051

Temporal Correlation - GLMM

Table 10: GLMM Temporal Correlation Model. Type I error rates and Power evaluated for each analytical and simulation method for theoretical residuals. Results are partitioned out by model mis-specification (from left to right) and residual type (top to bottom).

test	method	Type I Error	Power		
		Correct	A: iid	B: Gamma - Normal	C: Misp Cov
Anderson-Darling	Pearson	0.976	1.000	1.000	1.000
Anderson-Darling	one-step cdf	0.097	1.000	0.961	1.000
Anderson-Darling	one-step Generic	0.041	1.000	0.959	1.000
Anderson-Darling	MCMC	0.048	1.000	0.684	0.063
Anderson-Darling	Process osa	0.091	NA	0.919	1.000
Anderson-Darling	ecdf	0.227	1.000	1.000	1.000
Kolmogorov- Smirnov	Pearson	0.875	1.000	1.000	1.000
Kolmogorov- Smirnov	one-step cdf	0.077	1.000	0.874	1.000
Kolmogorov- Smirnov	one-step Generic	0.033	1.000	0.880	1.000
Kolmogorov- Smirnov	MCMC	0.056	1.000	0.581	0.055
Kolmogorov- Smirnov	Process osa	0.080	NA	0.793	1.000
Kolmogorov- Smirnov	ecdf	0.049	1.000	1.000	1.000
Autocorrelation	Pearson	0.217	1.000	1.000	1.000
Autocorrelation	one-step cdf	0.054	0.995	0.331	0.000
Autocorrelation	one-step Generic	0.053	0.993	0.092	0.000
Autocorrelation	MCMC	0.048	1.000	0.027	0.120
Autocorrelation	Process osa	0.051	NA	0.237	0.000
Autocorrelation	ecdf	0.051	0.975	1.000	0.986

Table 11: GLMM Temporal Correlation Model. Type I error rates and Power evaluated for each analytical and simulation method for estimated residuals. Results are partitioned out by model mis-specification (from left to right) and residual type (top to bottom).

test	method	Type I Error	Power		
		Correct	A: iid	B: Gamma - Normal	C: Misp Cov
Anderson-Darling	Pearson	0.619	0.911	0.001	1.000
Anderson-Darling	one-step cdf	0.115	0.830	0.289	0.025
Anderson-Darling	one-step Generic	0.039	0.886	0.020	0.020
Anderson-Darling	MCMC	0.043	0.841	0.057	0.051
Anderson-Darling	Process osa	0.069	NA	0.155	0.954
Anderson-Darling	ecdf	0.000	0.861	0.226	0.041
Kolmogorov- Smirnov	Pearson	0.878	0.986	0.982	1.000
Kolmogorov- Smirnov	one-step cdf	0.115	0.958	0.797	0.002
Kolmogorov- Smirnov	one-step Generic	0.000	0.965	0.795	0.002
Kolmogorov- Smirnov	MCMC	0.037	0.958	0.086	0.047
Kolmogorov- Smirnov	Process osa	0.088	NA	0.386	1.000
Kolmogorov- Smirnov	ecdf	0.924	0.951	0.979	0.040
Lilliefors	Pearson	0.993	0.983	0.966	1.000
Lilliefors	one-step cdf	0.184	0.779	0.956	0.126
Lilliefors	one-step Generic	0.049	0.779	0.961	0.123
Lilliefors	MCMC	0.036	0.779	0.162	0.048
Lilliefors	Process osa	0.055	NA	0.675	0.090
Lilliefors	ecdf	0.051	0.870	0.696	0.075
Autocorrelation	Pearson	0.005	1.000	0.021	0.034
Autocorrelation	one-step cdf	0.001	1.000	0.107	0.004
Autocorrelation	one-step Generic	0.000	1.000	0.122	0.005
Autocorrelation	MCMC	0.045	1.000	0.087	0.033
Autocorrelation	Process osa	0.008	NA	0.208	0.073
Autocorrelation	ecdf	0.000	0.999	0.036	0.000

Table 12: LMM Spatial Model. Type I error rates and Power evaluated for each analytical and simulation method for theoretical residuals. Results are partitioned out by model mis-specification (from left to right) and residual type (top to bottom).

test	method	Type I Error	Power		
		Correct	A: iid	B: Misp Data	C: Misp Cov
Anderson-Darling	Pearson	0.044	0.999	1.000	1.000
Anderson-Darling	one-step cdf	0.009	0.999	1.000	0.883
Anderson-Darling	one-step Generic	0.011	0.999	1.000	0.778
Anderson-Darling	one-step Gaussian	0.011	0.999	1.000	0.889
Anderson-Darling	full Gaussian	0.011	0.999	1.000	0.889
Anderson-Darling	MCMC	0.050	0.999	1.000	0.761
Anderson-Darling	ecdf	0.216	1.000	1.000	1.000
Kolmogorov- Smirnov	Pearson	0.046	0.992	0.977	1.000
Kolmogorov- Smirnov	one-step cdf	0.012	0.992	0.978	0.536
Kolmogorov- Smirnov	one-step Generic	0.012	0.992	0.973	0.538
Kolmogorov- Smirnov	one-step Gaussian	0.012	0.992	0.973	0.538
Kolmogorov- Smirnov	full Gaussian	0.012	0.992	0.973	0.538
Kolmogorov- Smirnov	MCMC	0.038	0.992	0.948	0.487
Kolmogorov- Smirnov	ecdf	0.039	0.992	0.980	1.000
Autocorrelation	Pearson	0.059	0.998	0.148	0.958
Autocorrelation	one-step cdf	0.050	0.998	0.027	0.175
Autocorrelation	one-step Generic	0.050	0.998	0.012	0.334
Autocorrelation	one-step Gaussian	0.050	0.998	0.011	0.314
Autocorrelation	full Gaussian	0.050	0.998	0.011	0.314
Autocorrelation	MCMC	0.043	0.998	0.001	0.147
Autocorrelation	ecdf	0.054	0.998	0.057	0.515

Table 13: LMM Spatial Model. Type I error rates and Power evaluated for each analytical and simulation method for estimated residuals. Results are partitioned out by model mis-specification (from left to right) and residual type (top to bottom).

test	method	Type I Error	Power		
		Correct	A: iid	B: Misp Data	C: Misp Cov
Anderson-Darling	Pearson	0.002	0.033	0.054	0.197
Anderson-Darling	one-step cdf	0.038	0.021	0.055	0.181
Anderson-Darling	one-step Generic	0.043	0.037	NA	NA
Anderson-Darling	one-step Gaussian	0.041	0.033	0.055	0.189
Anderson-Darling	full Gaussian	0.042	0.040	0.055	0.188
Anderson-Darling	MCMC	0.053	0.035	0.062	0.222
Anderson-Darling	ecdf	0.016	0.154	0.993	0.874
Kolmogorov- Smirnov	Pearson	0.139	0.004	0.985	0.935
Kolmogorov- Smirnov	one-step cdf	0.019	0.004	0.982	0.871
Kolmogorov- Smirnov	one-step Generic	0.013	0.004	NA	NA
Kolmogorov- Smirnov	one-step Gaussian	0.013	0.004	0.982	0.868
Kolmogorov- Smirnov	full Gaussian	0.013	0.004	0.981	0.896
Kolmogorov- Smirnov	MCMC	0.039	0.004	0.983	0.846
Kolmogorov- Smirnov	ecdf	0.146	0.002	0.986	0.940
Lilliefors	Pearson	0.054	0.124	0.999	0.916
Lilliefors	one-step cdf	0.054	0.124	0.999	0.943
Lilliefors	one-step Generic	0.048	0.122	NA	NA
Lilliefors	one-step Gaussian	0.048	0.124	0.999	0.947
Lilliefors	full Gaussian	0.048	0.124	0.998	0.954
Lilliefors	MCMC	0.067	0.124	0.999	0.860
Lilliefors	ecdf	0.064	0.172	0.999	0.927
Autocorrelation	Pearson	0.004	0.998	0.753	0.854
Autocorrelation	one-step cdf	0.045	0.998	0.737	0.859
Autocorrelation	one-step Generic	0.046	0.998	NA	NA
Autocorrelation	one-step Gaussian	0.046	0.998	0.738	0.861
Autocorrelation	full Gaussian	0.046	0.998	0.758	0.880
Autocorrelation	MCMC	0.038	0.998	0.753	0.864
Autocorrelation	ecdf	0.004	0.998	0.860	0.850

Table 14: GLMM Spatial Model. Type I error rates and Power evaluated for each analytical and simulation method for theoretical residuals. Results are partitioned out by model mis-specification (from left to right) and residual type (top to bottom).

test	method	Type I Error	Power		
		Correct	A: iid	B: Misp Data	C: Misp Cov
Anderson-Darling	Pearson	0.099	0.995	1.000	0.099
Anderson-Darling	one-step cdf	0.040	0.991	1.000	0.140
Anderson-Darling	one-step Generic	0.037	0.991	1.000	0.130
Anderson-Darling	MCMC	0.060	0.991	1.000	0.082
Anderson-Darling	ecdf	0.139	0.997	1.000	0.129
Kolmogorov- Smirnov	Pearson	0.085	0.979	1.000	0.085
Kolmogorov- Smirnov	one-step cdf	0.040	0.916	1.000	0.084
Kolmogorov- Smirnov	one-step Generic	0.037	0.916	1.000	0.083
Kolmogorov- Smirnov	MCMC	0.044	0.912	0.986	0.061
Kolmogorov- Smirnov	ecdf	0.052	0.912	1.000	0.057
Autocorrelation	Pearson	0.064	0.988	0.094	0.064
Autocorrelation	one-step cdf	0.055	0.990	0.000	0.620
Autocorrelation	one-step Generic	0.055	0.989	0.001	0.618
Autocorrelation	MCMC	0.071	0.988	0.000	0.265
Autocorrelation	ecdf	0.063	0.985	0.060	0.062

Table 15: GLMM Spatial Model. Type I error rates and Power evaluated for each analytical and simulation method for estimated residuals. Results are partitioned out by model mis-specification (from left to right) and residual type (top to bottom).

test	method	Type I Error	Power		
		Correct	A: iid	B: Misp Data	C: Misp Cov
Anderson-Darling	Pearson	0.005	0.829	0.007	0.003
Anderson-Darling	one-step cdf	0.027	0.751	0.221	0.049
Anderson-Darling	one-step Generic	0.036	0.759	0.129	0.038
Anderson-Darling	MCMC	0.049	0.753	0.096	0.045
Anderson-Darling	ecdf	0.009	0.970	0.005	0.006
Kolmogorov- Smirnov	Pearson	0.253	0.984	1.000	0.555
Kolmogorov- Smirnov	one-step cdf	0.006	0.820	0.931	0.001
Kolmogorov- Smirnov	one-step Generic	0.007	0.820	0.745	0.001
Kolmogorov- Smirnov	MCMC	0.038	0.820	0.089	0.022
Kolmogorov- Smirnov	ecdf	0.144	0.813	1.000	0.473
Lilliefors	Pearson	0.062	0.972	1.000	0.129
Lilliefors	one-step cdf	0.047	0.309	0.996	0.044
Lilliefors	one-step Generic	0.047	0.270	0.994	0.039
Lilliefors	MCMC	0.046	0.318	0.074	0.042
Lilliefors	ecdf	0.153	0.696	1.000	0.292
Autocorrelation	Pearson	0.001	0.988	0.001	0.348
Autocorrelation	one-step cdf	0.079	0.990	0.009	0.829
Autocorrelation	one-step Generic	0.079	0.989	0.008	0.824
Autocorrelation	MCMC	0.062	0.989	0.024	0.298
Autocorrelation	ecdf	0.001	0.989	0.004	0.335

Phylogenetic

Table 16: LMM Phylogenetic Model. Type I error rates and Power evaluated for each analytical and simulation method for theoretical residuals. Results are partitioned out by model mis-specification (from left to right) and residual type (top to bottom).

test	method	Type I Error	Power		
		Correct	A: iid	B: Misp Data	C: Misp Cov
Anderson-Darling	Pearson	0.055	1.000	1.000	1.000
Anderson-Darling	one-step cdf	0.052	1.000	0.980	0.989
Anderson-Darling	one-step Generic	0.047	1.000	0.924	0.937
Anderson-Darling	one-step Gaussian	0.047	1.000	0.986	0.994
Anderson-Darling	full Gaussian	0.047	1.000	0.986	0.994
Anderson-Darling	MCMC	0.042	1.000	0.916	0.905
Anderson-Darling	ecdf	0.229	1.000	1.000	1.000
Kolmogorov- Smirnov	Pearson	0.047	1.000	1.000	1.000
Kolmogorov- Smirnov	one-step cdf	0.043	1.000	0.738	0.750
Kolmogorov- Smirnov	one-step Generic	0.042	1.000	0.739	0.752
Kolmogorov- Smirnov	one-step Gaussian	0.042	1.000	0.740	0.752
Kolmogorov- Smirnov	full Gaussian	0.042	1.000	0.740	0.752
Kolmogorov- Smirnov	MCMC	0.042	1.000	0.758	0.737
Kolmogorov- Smirnov	ecdf	0.043	1.000	1.000	1.000
Autocorrelation	Pearson	0.045	0.999	0.850	0.872
Autocorrelation	one-step cdf	0.041	0.999	0.622	0.621
Autocorrelation	one-step Generic	0.042	0.999	0.321	0.326
Autocorrelation	one-step Gaussian	0.042	0.999	0.434	0.442
Autocorrelation	full Gaussian	0.042	0.999	0.434	0.442
Autocorrelation	MCMC	0.051	0.999	0.337	0.329
Autocorrelation	ecdf	0.051	0.997	0.354	0.363

Table 17: LMM Phylogenetic Model. Type I error rates and Power evaluated for each analytical and simulation method for estimated residuals. Results are partitioned out by model mis-specification (from left to right) and residual type (top to bottom).

test	method	Type I Error	Power		
		Correct	A: iid	B: Misp Data	C: Misp Cov
Anderson-Darling	Pearson	0.000	0.034	0.221	0.201
Anderson-Darling	one-step cdf	0.068	0.038	0.610	0.610
Anderson-Darling	one-step Generic	0.037	0.034	0.016	0.006
Anderson-Darling	one-step Gaussian	0.029	0.029	0.447	0.430
Anderson-Darling	full Gaussian	0.028	0.028	0.451	0.436
Anderson-Darling	MCMC	0.038	0.038	0.216	0.215
Anderson-Darling	ecdf	0.003	0.151	0.530	0.496
Kolmogorov- Smirnov	Pearson	0.638	0.005	0.991	0.993
Kolmogorov- Smirnov	one-step cdf	0.046	0.005	0.979	0.972
Kolmogorov- Smirnov	one-step Generic	0.013	0.005	0.987	0.981
Kolmogorov- Smirnov	one-step Gaussian	0.013	0.005	0.987	0.981
Kolmogorov- Smirnov	full Gaussian	0.013	0.005	0.986	0.980
Kolmogorov- Smirnov	MCMC	0.043	0.005	0.446	0.416
Kolmogorov- Smirnov	ecdf	0.629	0.005	0.992	0.992
Lilliefors	Pearson	0.049	0.122	0.995	0.994
Lilliefors	one-step cdf	0.075	0.122	0.987	0.987
Lilliefors	one-step Generic	0.042	0.122	0.993	0.991
Lilliefors	one-step Gaussian	0.042	0.122	0.997	0.995
Lilliefors	full Gaussian	0.042	0.122	0.997	0.995
Lilliefors	MCMC	0.044	0.122	0.524	0.474
Lilliefors	ecdf	0.048	0.201	0.605	0.578
Autocorrelation	Pearson	0.935	0.999	0.798	0.805
Autocorrelation	one-step cdf	0.019	0.999	0.391	0.387
Autocorrelation	one-step Generic	0.021	0.999	0.473	0.478
Autocorrelation	one-step Gaussian	0.021	0.999	0.484	0.488
Autocorrelation	full Gaussian	0.021	0.999	0.484	0.485
Autocorrelation	MCMC	0.035	0.999	0.292	0.278
Autocorrelation	ecdf	0.902	0.999	0.424	0.407

Table 18: GLMM Phylogenetic Model. Type I error rates and Power evaluated for each analytical and simulation method for theoretical residuals. Results are partitioned out by model mis-specification (from left to right) and residual type (top to bottom).

test	method	Type I Error	Power		
		Correct	A: iid	C: Misp Cov	NA
Anderson-Darling	Pearson	0.531	1.000	1.000	1.000
Anderson-Darling	one-step cdf	0.112	1.000	0.971	0.234
Anderson-Darling	one-step Generic	0.023	1.000	0.969	0.094
Anderson-Darling	MCMC	0.049	1.000	0.271	0.059
Anderson-Darling	ecdf	0.091	1.000	1.000	0.964
Kolmogorov- Smirnov	Pearson	0.536	1.000	1.000	0.928
Kolmogorov- Smirnov	one-step cdf	0.098	0.998	0.948	0.140
Kolmogorov- Smirnov	one-step Generic	0.026	0.997	0.926	0.052
Kolmogorov- Smirnov	MCMC	0.044	0.997	0.230	0.041
Kolmogorov- Smirnov	ecdf	0.031	0.999	1.000	0.597
Autocorrelation	Pearson	0.041	0.976	0.952	0.157
Autocorrelation	one-step cdf	0.056	0.984	0.308	0.090
Autocorrelation	one-step Generic	0.039	0.989	0.316	0.085
Autocorrelation	MCMC	0.042	0.991	0.084	0.064
Autocorrelation	ecdf	0.051	0.983	0.905	0.059

Table 19: GLMM Phylogenetic Model. Type I error rates and Power evaluated for each analytical and simulation method for estimated residuals. Results are partitioned out by model mis-specification (from left to right) and residual type (top to bottom).

test	method	Type I Error	Power		
		Correct	A: iid	C: Misp Cov	NA
Anderson-Darling	Pearson	0.004	0.998	0.012	0.003
Anderson-Darling	one-step cdf	0.079	0.985	0.027	0.101
Anderson-Darling	one-step Generic	0.041	0.983	0.022	0.040
Anderson-Darling	MCMC	0.056	0.985	0.047	0.052
Anderson-Darling	ecdf	0.001	1.000	0.031	0.001
Kolmogorov- Smirnov	Pearson	1.000	1.000	0.150	1.000
Kolmogorov- Smirnov	one-step cdf	0.061	0.996	0.017	0.091
Kolmogorov- Smirnov	one-step Generic	0.012	0.996	0.018	0.014
Kolmogorov- Smirnov	MCMC	0.028	0.996	0.033	0.055
Kolmogorov- Smirnov	ecdf	0.855	0.995	0.077	0.919
Lilliefors	Pearson	0.662	1.000	0.164	0.826
Lilliefors	one-step cdf	0.059	0.962	0.076	0.068
Lilliefors	one-step Generic	0.024	0.967	0.081	0.030
Lilliefors	MCMC	0.050	0.988	0.063	0.054
Lilliefors	ecdf	0.796	0.999	0.131	0.865
Autocorrelation	Pearson	0.621	0.975	0.240	0.863
Autocorrelation	one-step cdf	0.063	0.967	0.158	0.091
Autocorrelation	one-step Generic	0.048	0.985	0.160	0.097
Autocorrelation	MCMC	0.051	0.988	0.068	0.055
Autocorrelation	ecdf	0.360	0.977	0.224	0.530

Table 20: Overview of issues and recommendations for common classes of models. Correlation and distributions refer to predicted data from a fitted model, against which observed points are compared. A linear rotation refers to a multiplication of the simulated and observed data by a Cholesky decomposition of the estimated covariance matrix of the observed data, $z'=Lz$, as available in DHARMa.

Model class		Case studies	Issues and causes	Recommendation
Linear model		Linear model	No issues	Pearson residuals
Generalized linear model (GLM)	linear	Skewed Gamma	Non-normality resulting from response variable. Quantile residuals are needed if not approximately normal.	Quantile residual
Linear mixed model (LMM), Multivariate model		Random walk, Spatial LMM, Multinomial	Linear correlations caused by non-independence in observations.	Use a method that linearly decorrelates in order to transform to a unit iid normal. OSA Full Gaussian, OSA one-step Gaussian, or simulation residuals with rotation.
Generalized linear mixed model (GLMM)	linear	Spatial Poisson, Repeated measures Tweedie	Non-normality and non-linear correlations caused by response variable and non-independence in observations.	Needs non-linear decorrelation and quantiles. Needs non-linear decorrelation quantiles. Best approach depends on study and sample size.